# Aesha Gandhi

919-520-1149 | aeshagandhi27@gmail.com | linkedin.com/in/aesha-gandhi | github.com/aeshagandhi

## Education

**Duke University** Durham, NC
*Master's in Data Science* GPA: 3.962 *Aug 2025 – May 2027 (Expected)*
- Coursework: Agentic AI, Deep Learning, Reinforcement Learning, Explainable AI, Statistics, Data Engineering, NLP, ML

**University of California, Berkeley** Berkeley, CA
*Bachelor of Arts in Data Science* GPA: 3.72 *Aug 2021 – May 2025*
- Domain emphasis in Applied Mathematics and Modeling; Minors in Computer Science, Industrial Engineering
- Coursework: Artificial Intelligence, Algorithms, Linear Programming, Inference/Decisions, Data Mining, Discrete Math, Probability Theory, Data Ethics, Data Structures, Database Systems

**North Carolina School of Science and Math (NCSSM)** Durham, NC

## Experience

**SAS** Cary, NC
*Data Science Intern, Computer Vision* (Year-Round) *May 2026 – Present*
- Develop computer vision solutions within the Applied AI and Modeling division, currently fine-tuning an object detection model for PPE detection in industrial warehouse environments
- Developed and optimized an RF-DETR instance segmentation model in Python for a worker-safety pipeline, achieving 0.91 mAP50–95 and 0.90 mean IoU with MLflow-based training and evaluation on NVIDIA A100 GPUs

*Applied AI Engineering Intern* (Year-Round) *May 2025 – May 2026*
- Built production-ready Python and TypeScript SDKs for containerized ML model deployment and management, leveraging OpenAPI-based code generation, testing, packaging, and CI/CD integration
- Developed the analytics foundation for a centralized engineering metrics platform by building SQL and Python workflows to collect KPIs from large-scale DevOps data and automate Python and RPM dependency reporting

*Interactive Analytics and Modeling Intern* (Year-Round) *May 2022 – May 2025*
- Designed, implemented, and shipped a hyperparameter tuning results feature for SAS Viya Visual Analytics used by 700+ enterprise customers, building React-based front-end dialogs and visualizations that surfaced training progress, objective history, and model performance across machine learning and statistical models

**College of Computing, Data Science, and Society** Berkeley, CA
*Data Science Course Tutor* *Jan. 2024 – May 2024*
- Graded 250+ students' work and led weekly office hours for a course in probabilistic inference and decision making

## Research & Applied AI

**Duke AI Trust Lab** | *Researcher, Multi-Agent AI Safety* Sep 2026 – Present
- Research emergent deception and misalignment in multi-agent AI systems through controlled simulations in a Catan negotiation environment, measuring contradictions between agents' public statements and environment-recorded actions

**Bank of New York & Duke University** | *Data Science Capstone, Agentic AI* Aug 2026 – Present
- Develop an agentic AI workflow for corporate action intelligence with Bank of New York, focused on extracting, validating, and normalizing events from financial notices and external data sources while presenting weekly progress to the BNY AI Hub team and stakeholders

## Projects

**Intelligent Credit Card Advisor Agent** | *LLMs, OCR, Transformer Classifiers, Streamlit, Python* Spring 2026
- Developed an agentic receipt to recommendation pipeline that extracts spending patterns from receipt images using CV and OCR (Tesseract, PaddleOCR), classifies line items with a transformer based classifier, and leverages LLM powered web search tool use to generate grounded, real time credit card recommendations

**Codenames Spymaster RL Agent** | *PyTorch, Sentence Transformers, Python* Spring 2026
- Designed a goal-conditioned RL agent for the Codenames spymaster task using sentence-transformer embeddings for semantic board representation, benchmarking Greedy, Behavioral Cloning, and Soft Actor-Critic with Hindsight Experience Replay pipelines to study imitation warm-starts and reward shaping for semantic clue generation

**Vision-Language Image Captioning** | *Hugging Face Transformers, PyTorch* Fall 2025
- Reversed the text-to-image generation paradigm by fine-tuning vision-language models (ViT-GPT2, BLIP) on 10,000 DiffusionDB image–caption pairs, achieving a 102% improvement in combined BLEU/BERTScore on NVIDIA A100 GPUs

**AI Health Companion** | *LLMs, AI Agents, Real-Time Systems* Hackathon Project
- Built a voice-first AI companion for seniors using OpenAI's Realtime API over WebSocket for low-latency audio, with a multi-layered prompt pipeline that converts conversation into structured assessments across 5 health dimensions with weighted severity scoring, surfaced via a family dashboard

## Technical Skills

**Languages**: Python, SQL, Java, JavaScript/TypeScript, R, SAS Base
**Frameworks**: PyTorch, Transformers, Scikit-learn, LLMs, AI Agents, Agentic Workflows, RAG, Fine-Tuning, Tool Calling, TensorFlow, Apache Spark/PySpark, Prompt Engineering, Apache Airflow/Kafka, React
**Developer Tools**: MLflow, Docker, OpenAI APIs, Databricks, AWS/GCP/Azure, Postgres, MongoDB, Git/GitHub
**Libraries**: OpenCV, Sentence-Transformers, Pandas/Polars, NumPy, Matplotlib